of the Open Source OLAP Systems for Big Data: ClickHouse, Druid, and Pinot," by Roman Leventov (http://mng.bz/9DRx).

In this context, it's interesting to note that columnar storage is not limited to backends. For example, in OpenTelemetry there is ongoing work to introduce columnar encoding (https://github.com/open-telemetry/oteps/pull/171) in the collector and protocol, to speed up processing and improve performance and scalability.

Tip

The space of columnar data stores and databases, formats, and open source projects has exploded in the past decade. In fact, my first industry job (after some 12 years in research) was in the context of a Hadoop start-up on the now-retired Apache Drill project (which was similar to the open source version of Google's Dremel, which forms the basis of BigQuery).

There are many interesting data stores that you may be interested in, including MonetDB, Hypertable, Apache Kudu, and Greenplum. Covering them in any detail goes beyond the scope of this book. If you want to learn more, I recommend reading Martin Kleppmann's excellent book *Designing Data-Intensive Applications* (O'Reilly, 2017; https://dataintensive.net/) and *Fundamentals of Data Observability* (O'Reilly, 2023; http://mng.bz/jPQz) by the one and only Andy Petrella.

After that much theory, we deserve some action! So let's dive deep into an example with an open source columnar data store that gained a lot of traction in the past few years: ClickHouse.

In this walkthrough, you will (again) use the `ho11y` generator, configured to send its logs via the Fluentd logs driver to the OpenTelemetry collector with a pipeline using the Fluent Forward receiver (http://mng.bz/WzD4), and the ClickHouse exporter (http://mng.bz/8rZZ) to ingest the logs into ClickHouse, where you then will be able to query the logs using SQL.

Excited? Let's go!

We start off with the Docker Compose configuration shown in the next listing. You know most of the setup already from previous examples, so we will focus on the new parts of this setup.

Listing 5.1 Snippets of the Docker Compose configuration for the ClickHouse setup

```
ho11y:
    image: public.ecr.aws/mhausenblas/ho11y:stable
    ports:
      - 8765:8765
    logging:
      driver: fluentd
      options:
        fluentd-address: "localhost:8006"

  otel-collector:
    image: otel/opentelemetry-collector-contrib:0.55.0
```

Here, we provide the Docker Fluentd driver with the address of the fluentforward receiver (part of the OpenTelemetry collector pipeline).